10 SECONDARY STORAGE

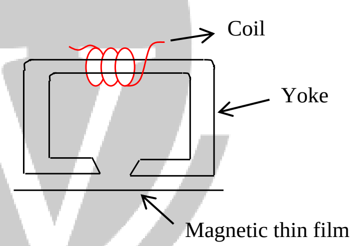
10.1 Introduction

- The cost per bit of stored information is high in c semi-conductor memories (main memory). This limits the use of these memories for large storage. Large storage requirements of most of computers are economically fulfilled by magnetic disks, magnetic tapes and optical disk memories. These memories are referred as secondary storage devices.

10.2 Magnetic Disks

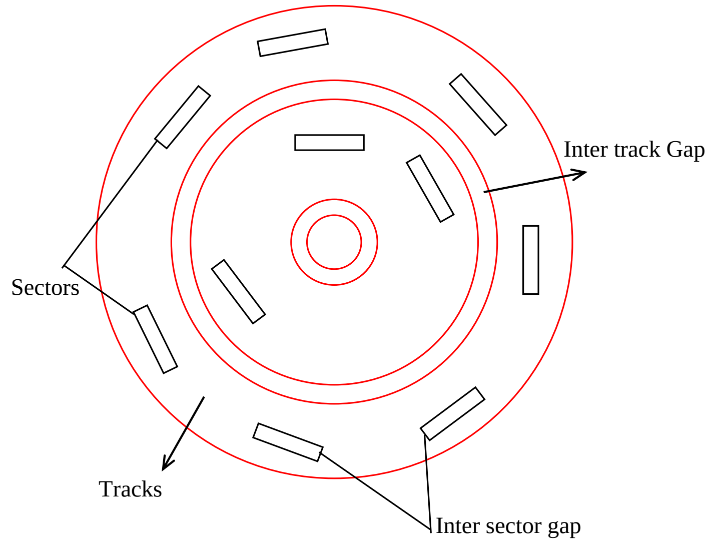
- A magnetic disk is a thin, circular metal plate. It is coated with a thin magnetic film usually on both sides.
- Digital information is stored on the magnetic disk by magnetizing the magnetic surface in a particular direction.
- Magnetic disks are semi random access memories.

The head consists of a magnetic yoke and the magnetizing coil. The Digital information can be stored on the film by applying current pulses of suitable polarity.



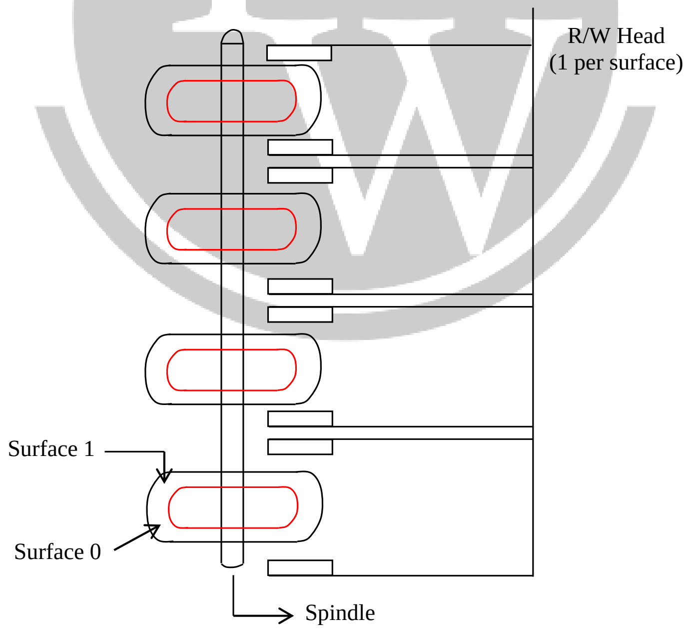
10.2.1. Data organization & Formatting

- The data on the disk is organized in a concentration set of rings. These concentric set of rings are called tracks.
  - ✓ Each track has same width as head and adjacent tracks are separated by gaps.
  - ✓ Each track is divided into manageable units called sectors.
  - ✓ Each sector stores a block of data which can be transferred to or from the disk.
  - ✓ The universal size of sector is 512 bytes.
  - ✓ The R/W head is capable of reading a sector from disk and writing a sector to disk.
  - ✓ Placing R/W head on desired track is random access and reading concerned sector is serial. Hence disks are semi random access devices.



10.2.2. Physical Characteristics

(i)	Head Motion	-	Fixed head (one per track)
			Movable head (one per surface)
(ii)	Disk portability	-	Non removable or Removable
(iii)	Sides	-	Single sided, Double – sided.
(iv)	Disk/surface	-	Single surface, Multiple – surfaces
(v)	Head Mechanism	-	Contact, fixed Gap, Aerodynamic Gap.



A vertical set of all of the tracks with the same number on each surface of a diskette or hard disk is called "cylinder."

(vi) Platters - Single platter,
Multiple platters – some disk
Accommodate multiple platters
Stacked vertically a fraction of an inch apart.

(1) If all the tracks are having constant capacity then the disk moves with "constant angular capacity" and "recording density" varying from track to track.

(2) If each track is having variable capacity then the disk is moving with "constant linear velocity"

- Placing the control information in the sector is called as formatting.
- The disk controller is the hardware interface to control the operation of a disk drive. Used to control more than one drive. The major functions are seek, read, write and error checking.
- The start and end of each sector is determined by the control data stored in the sector.
- Disk performance parameters:

Seek Time: It is the time required to more the disk arm to the required track.

Rotational Delay: The time taken for the beginning of sector to reach. (latency)

Access time: The sum of seek time and the rotational delay.

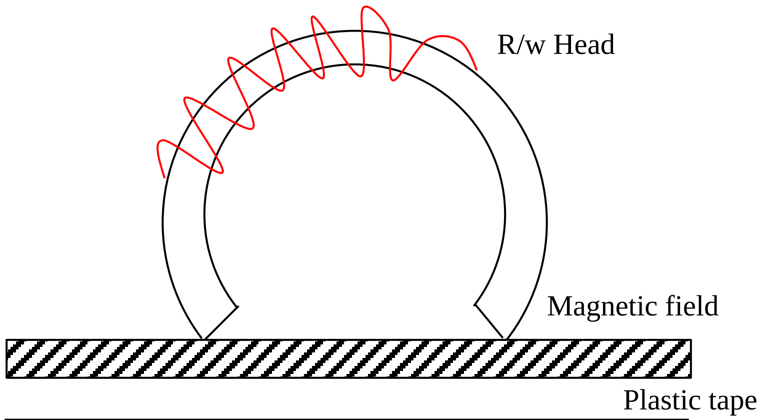
- The average time to disk access is

  $t_{avg} = t_{seektime} + t_{rotational\ delay} + t_{data\ transfer} + t_{over\ head}$
- Maximum Recording Density (P) = No. of Bytes/cm
- Data transfer rate (D) = Number of Bytes/Sec.

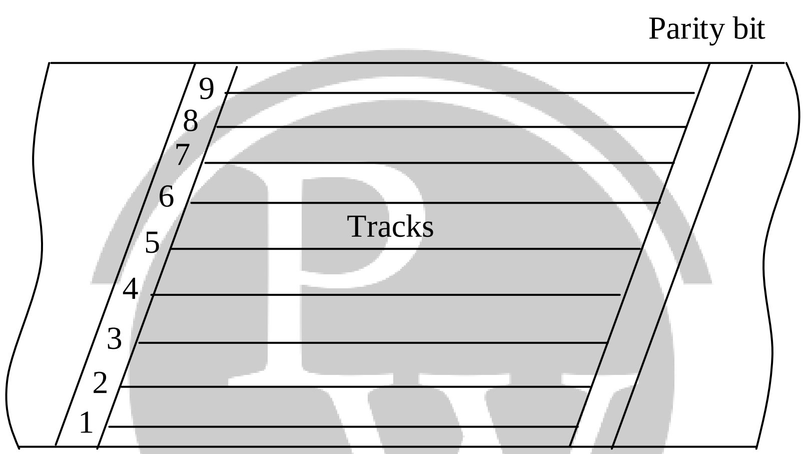
10.3 Secondary Storage Devices

10.3.1. Magnetic Tapes

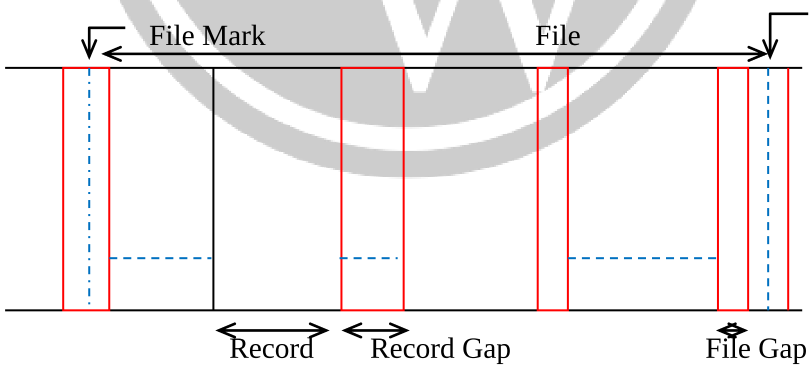
- Magnetic tape is one of the most popular storage medium for large data that are sequentially accessed and processed.
- The tape is formed by depositing magnetic film on a very thin and ½ or ¼ inch wide plastic tape.
- Usually, iron oxide is used as a magnetizing material.
- The information is recorded on the tape with the help of read/write head. It magnetizes or non magnetizes tiny invisible spots (1's & 0's).
- Usually 7 or a bits are recorded (a character) in parallel across the width of the tape, perpendicular to the direction of motion. A separate R/Wo head is provided for each bit position on the tape.



- Data on the tape are organized in the form of records separated by gaps.
- A set of related records constitutes a file.



- The data on a 9^{th} track (a) tape is stored in parallel consists of data byte and a parity bit.



- Data transfer takes place when the tape is moving at constant velocity relative to a read/write head.
- Hence the maximum data transfer rate depends largely on the storage density along the tape and speed of the tape.
- The file mark is used as header or identifier.
- The information on the magnetic tape is organized into blocks. These blocks have a fixed length and separated by gap between them.
- If the block length is B_L and inter block gap length is G_L, then the utilization factor of the tape (u) is given by

$$u = \frac{B_L}{B_L + G_L}$$

- The unit of data transfer is a record.
- The tape is moving with a linear velocity and Read/write head is constant.
- Let L is the length of the tape,
  - N is the number of parallel tracks,
  - P is the constant recording density.

(i) Capacity of the tape

$$C = L \times N \times P$$

(ii) Let V is the linear velocity of the tape, then the data transfer rate

$$D = V * N * P$$

✓ With utilization factor, the effective data transfer rate is

$$D_{eff} = D * \frac{B_L}{B_L + G_L}$$

✓ Due to inter block gap and time needed to start and stop and tape between accesses, the effective data transfer rate d_{eff} is actually less than the maximum rate d.

$$d_{eff} = \frac{t_D \times d}{t_D + t_G + t_{ss}}$$

t_D = time to scan a data block

t_G = time to scan the inter block gap.

T_{ss} = time to start and stop the tape.

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